HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · HOME TEXTILES · QUALITY & PRODUCT DEVELOPMENT

TOWELS · BED LINEN · RUGS · SHIPPED TO GLOBAL RETAILERS

Colour Extraction & Shade Matching

A colour studio for the quality and product development teams: it reads every colour in a design, scores each print run against the approved standard with ΔE2000, and shade-bands runs before they are packed.

~~2.3~~ → **0.9** average ΔE2000, run to standard



64 % fewer
Shade-related shipment rejections
▼ caught at the run, not the buyer



0.9 ΔE2000
Average run-to-standard difference
▼ from about 2.3



10 minutes
To compare a run with the standard
▼ from about 2 days



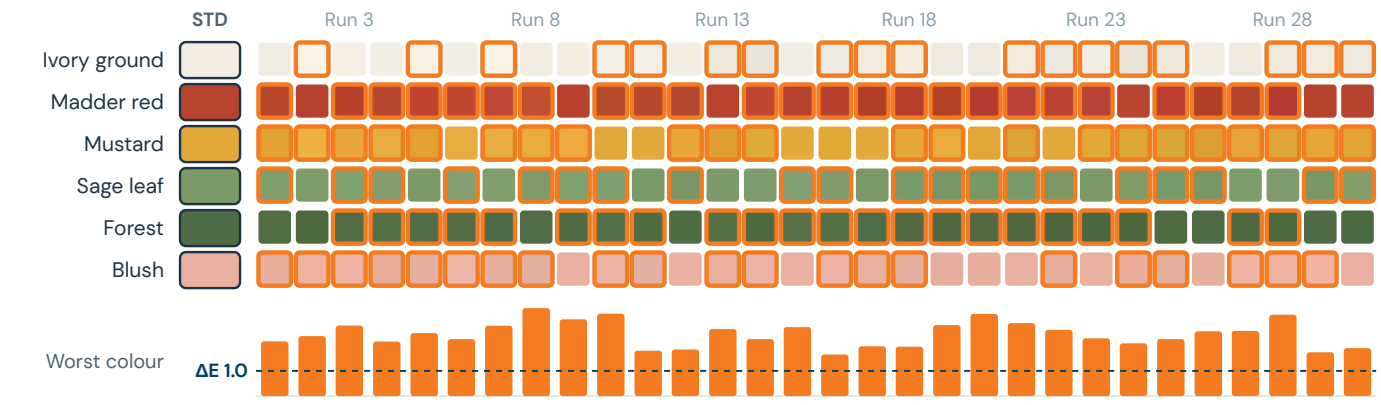
100 %
Runs shade-banded before packing
▲ no mixed shades in a shipment

Rings: share of rejections removed; ΔE now against the 2.3 it was (inner ring = before); 10 minutes against 2 days (inner ring = before); every run sorted into one of five shade bands. Figures indicative; validated metrics available on request.

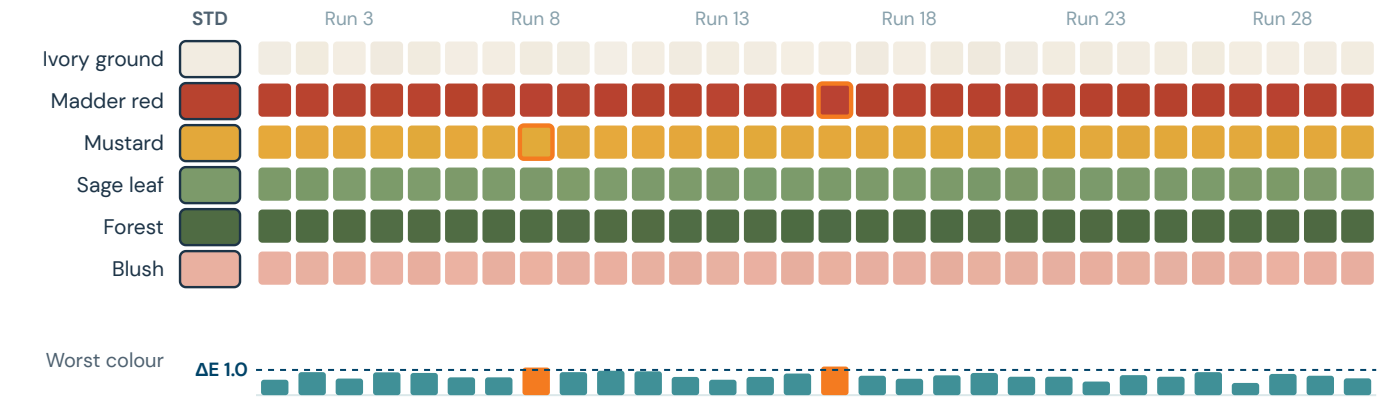
01 Thirty runs of one design, colour by colour

Each cell is the run's actual colour · orange outline = ΔE over 1.0 · illustrative, simulated runs

BEFORE approval by eye under a light box



AFTER every run scored against the standard and banded



Colour over ΔE 1.0 · Run's worst colour over 1.0 · Worst colour within 1.0 · STD = approved standard

BEFORE
30 of 30 runs
had at least one colour over ΔE 1.0: 127 of 180 swatches. Mean worst colour 2.51.

AFTER
2 of 30 runs
had a colour over ΔE 1.0: 2 of 180 swatches. Mean worst colour 0.79.

Why it matters: a run that looks fine under shop-floor lights can be visibly different next to last month's run on a retailer's shelf. Small drifts across machines, dye lots and weeks add up.

Read it like a colourist: the studio also says which way a shade moved (lighter, duller, redder), so the dye house knows what to fix.

02 One colour language, one agreed number

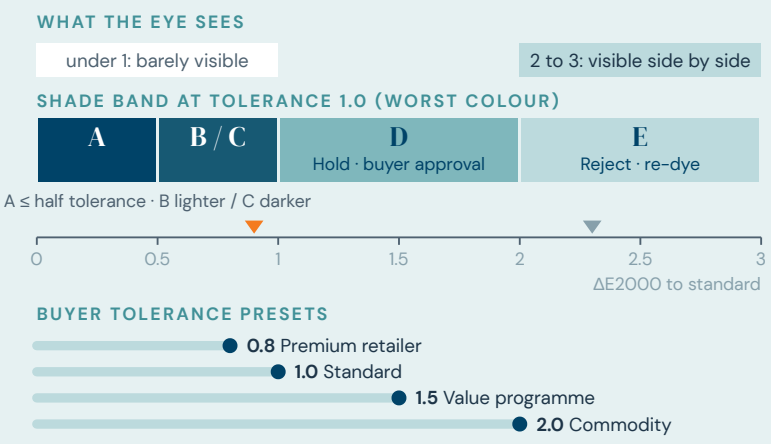
Every colour in every format · scored with ΔE2000 · banded before packing

COLOUR	HEX	RGB	HSB	HSL	CMYK	CIELAB
Ivory ground	#F2ECE1	242, 236, 225	39°, 7%, 95%	39°, 40%, 92%	0, 2, 7, 5	93.6, 0.1, 6.0
Madder red	#B8432F	184, 67, 47	9°, 74%, 72%	9°, 59%, 45%	0, 64, 74, 28	44.8, 46.2, 36.8
Mustard	#E3A83A	227, 168, 58	39°, 74%, 89%	39°, 75%, 56%	0, 26, 74, 11	72.7, 11.7, 62.1
Sage leaf	#7C9A6A	124, 154, 106	98°, 31%, 60%	98°, 19%, 51%	19, 0, 31, 40	60.3, -19.6, 21.9
Forest	#4F6B43	79, 107, 67	102°, 37%, 42%	102°, 23%, 34%	26, 0, 37, 58	42.1, -18.5, 19.2
Blush	#E9B0A0	233, 176, 160	13°, 31%, 91%	13°, 62%, 77%	0, 24, 31, 9	76.6, 18.6, 16.3

The six colours of the design in section 01, converted with the studio's own maths. Designers think in HEX and CMYK, dye labs in LAB, buyers in their own references; the studio carries all of them with **CIELAB as the reference space**. CMYK is a simple device conversion (full black), not an ICC press profile; LAB uses D65 and the 2° observer. In production, readings come from a spectrophotometer, not a photo.

How far is too far?

▼ Average run before: about 2.3 ▼ After: 0.9



03 Four pain points, four fixes

Before → after

BEFORE
Judged by eye
Approval relied on experienced eyes under a light box; two checkers could disagree.

ΔE2000 TOLERANCE

AFTER
A number, with direction
Each colour scored against the standard to the buyer's limit, with the way it moved.

BEFORE
Many runs, one standard
Same design across machines, dye lots and weeks; runs checked one by one, if at all.

SCORE EVERY RUN

AFTER
Every run scored
Spectrophotometer LAB readings uploaded per run and compared in minutes.

BEFORE
Mixed shades in a carton
Slightly different runs packed together and flagged by the buyer.

FIVE SHADE BANDS

AFTER
One band per shipment
The band is printed on the carton label; approval notes go to buyers with the numbers.

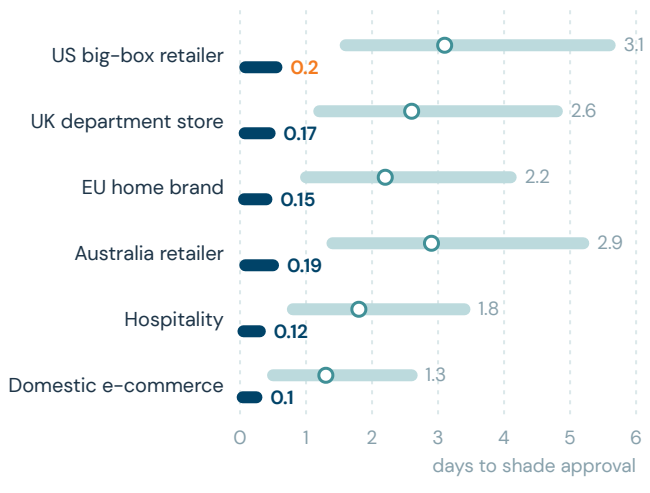
BEFORE
Five colour formats
Colours picked by hand and retyped between design, lab and buyer.

ONE COLOUR LANGUAGE

AFTER
Full palette in seconds
HEX, RGB, HSB, HSL, CMYK and LAB together, with each colour's share of the design.

04 Days to shade approval

Range and median by buyer programme · before (light), after (navy) · illustrative



05 New colourways

Same artwork, recoloured in CIELAB, keeping shading and texture · studio presets



06 How HMC ran it

2 weeks	**Discover** Rejected shipments traced to runs; deep reds, navies and greys drift most.
2 weeks	**Design** CIELAB reference, ΔE2000 with buyer tolerances, five shade bands.
5 weeks	**Build** Palette extraction, run matching with drift direction, colourway studio.
Pilot bed linen	**Adopt** Spectro first, studio second; bands on the carton label.
All categories	**Scale** Towels and rugs; shared library; drift alerts per machine and dye lot.

What any textile maker can borrow

01
Agree the number, not the look
A ΔE tolerance agreed with the buyer ends most shade arguments before they start.

02
Say which way it moved
"Lighter and redder" tells the dye house what to fix; a pass/fail alone does not.

03
Band before you pack
Even passing runs differ slightly; packing by band keeps every carton consistent.

Built with **CIELAB & CIEDE2000** · **k-means** · **Spectrophotometer data** · **OpenAI GPT** · **Power BI** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

HMC ONE
AI STRATEGY. BUILT TO SCALE.